

In theory, after the infinity of 1s and plus signs and fraction bars represented by the ellipsis, we should insert a phi into [Listing 5-2](#), just like it appears in the bottom right of [Listing 5-1](#). But we will never get through all of those 1s (because there are an infinite number of them), so we are justified in forgetting entirely about the phi that's supposed to be nested in the righthand side.

More about Continued Fractions

The expressions just shown are called continued fractions. A *continued fraction* consists of sums and reciprocals nested in multiple layers. Continued fractions can be finite, like the one in [Listing 5-1](#) that terminated after seven layers, or infinite, continuing forever without end like the one in [Listing 5-2](#). Continued fractions are especially useful for our purposes because they enable us to express the exact value of phi without needing to chop down an infinite forest to manufacture enough paper. In fact, mathematicians sometimes use an even more concise notation method that enables us to express a continued fraction in one simple line. Instead of writing all the fraction bars in a continued fraction, we can use square brackets ([]) to denote that we're working with a continued fraction, and use a semicolon to separate the digit that's "alone" from the digits that are together in a fraction. With this method, we can write the continued fraction for phi as the following:

$$\phi = [1; 1, 1, 1, 1 \dots]$$

In this case, the ellipses are no longer losing information, since the continued fraction for phi has a clear pattern: it's all 1s, so we know its exact 100th or 1,000th element. This is one of those times when mathematics seems to deliver something miraculous to us: a way to concisely write down a number that we had thought was infinite, without pattern, and ineffable. But phi isn't the only possible continued fraction. We could write another continued fraction as follows:

$$\text{mysterynumber} = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots]$$

In this case, after the first few digits, we find a simple pattern: pairs of 1s alternate with increasing even numbers. The next values will be 1, 1, 10, 1,